

Parv Kapoor

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Education

Carnegie Mellon University

Ph.D. in Software Engineering | Advisors: Dr. Eunsuk Kang and Dr. Sebastian Scherer

Pittsburgh, PA

May 2026 (expected)

Carnegie Mellon University

Masters in Software Engineering | TA: 10-623 Generative AI, 17-723 Designing large scale systems

Pittsburgh, PA

December 2025 (expected)

Manipal Institute of Technology

B.Tech. in Computer Science Engineering

Manipal, India

August 2020

Thesis: Specification-Driven Foundation Models for Safe Autonomous Systems

Skills

Programming

Python, C, C++, CUDA programming, JAVA, MATLAB, Alloy, TLA+, MySQL, OpenCL

Tools and Libraries

PyTorch, TensorFlow, ROS, Docker, Habitat, Isaac Gym

Experience

Microsoft Research

Machine Learning Research Intern | PI: Dr. John Langford and Dr. Siddhartha Sen

New York, NY

May 2025 – August 2025

- Built Matryoshka Belief State Transformer (MatBST) architecture with hierarchical latent planning, enabling fast long-horizon reasoning in LLMs
- Optimized large-scale pretraining and fine tuning with PyTorch FSDP and custom sharding strategies for large language models

General Robotics

Student Researcher (Part time) | PIs: Dr. Jonathan Huang and Dr. Ashish Kapoor

Seattle, WA

September 2024 - May 2025

- Developed pipelines for distributed synthetic data generation, pretraining and fine tuning of robot foundation models
- Developed aligned robotics foundation models using novel fine tuning paradigms for navigation and instruction following
- Contributing to the development of the AirGen simulation platform for aerial, ground, and manipulator robots

General Robotics

Research Intern | PIs: Dr. Sai Vemprala and Dr. Ashish Kapoor

Seattle, WA

May 2024 - August 24

- Developed novel pretraining methods for safe navigation foundation models, outperforming baselines by 74.3%.
- Built a foundation model field deployment pipeline for diverse robots (Unitree Go2, DJI Robomaster) using domain adaptation.
- Integrated multimodal foundation models into the GRID platform, developed and deployed skills chaining the models.

Cyber Physical Systems Lab, University of Southern California

Research Intern | PI: Dr. Jyotirmoy Vinay Deshmukh

Los Angeles, U.S.A.

January 2020 - January 2021

- Developed novel model-based reinforcement learning algorithms for safe policy training from signal temporal logic specifications
- Implemented efficient model-free algorithms (TRPO, A3C, PPO) in PyTorch with unique STL-based reward design
- Achieved 82% higher specification satisfaction compared to baseline RL policies
- Engineered in-house simulation environments for algorithm benchmarking employing CARLA, AirSim, and Gazebo

Visual Computing Group, Cardiff University

Research Intern | PI: Dr. David Marshall

Cardiff, U.K.

May 2019 - July 2019

- Constructed a safe trajectory prediction system for visually impaired individuals using ZED stereo camera
- Implemented and trained a 2-stream CNN in TensorFlow on human walking data for forecasting ego agent camera movement
- Improved CNN accuracy in low-data regimes through neuro-inspired data augmentation

Ph.D. Research

AirLab, Carnegie Mellon University

PI: Sebastian Scherer

Pittsburgh, PA

August 2021 – Ongoing

- Built constrained decoding frameworks for robotic transformers using formal logic over token spaces, outperforming baselines by 25%
- Pretrained LfD policies and refined them online via symbolic tree search, improving constraint satisfaction by 60% [ICRA'23]
- Developed vision-based control barrier functions for UAV safety, validated via sim-to-real transfer from Isaac Sim to field, achieving 71% improvement over baselines across 70+ flight hours [ICRA'22, RSS'25]

- Built STL_{CG++}, a PyTorch/JAX library for differentiable STL, enabling 100× faster robustness computation via attention masking [RA-L 2025]
- Trained world models and used pretrained visual embeddings (DINOv2, CLIP) with MPPI for logic-constrained object navigation
- Learned formal behavior rules from robot demonstrations using MaxSAT for structured policy synthesis [ICSE'25]
- Designed stochastic optimization methods to analyze RL policy robustness under environmental deviations [FM'24]
- Devised requirement decomposition for STL-constrained trajectory optimization, reducing solve time by 65% [NFM'24]

Preprints and Publications

Scaling Long-Horizon Reasoning via Hierarchical Matryoshka Belief State Transformers

P. Kapoor, S. Sen, J. Langford

- Under Submission

Constrained Decoding for Robotics Foundation Models

P. Kapoor, A. Ganlath, C. Liu, S. Scherer, E. Kang

- Under Submission [\[arxiv\]](#)

Pretrained Embeddings as a Behavior Specification Mechanism

P. Kapoor, A. Hammer, A. Kapoor, K. Leung, E. Kang

- Under Submission [\[arxiv\]](#)

ViSafe: Vision-enabled Safety for High-speed Detect and Avoid

P. Kapoor, I. Higgins, N. V. Keetha, J. Patrikar, I. Cisneros, Z. Ye, Y. He, Y. Hu, C. Liu, E. Kang, S. Scherer

- Robotics Science and Systems (RSS) 2025 [\[arxiv\]](#)

STL_{CG++}: A Masking Approach for Differentiable Signal Temporal Logic Specification

P. Kapoor, K. Mizuta, E. Kang, K. Leung

- IEEE Robotics and Automation Letters (RA-L), presented at IEEE International Conference on Robotics and Automation (ICRA) 2026 [\[arxiv\]](#)

Logically Constrained Robotics Transformers for Enhanced Perception Action Planning

P. Kapoor, S. Vemprala, A. Kapoor

- Robotics Science and Systems (RSS): Towards Safe Autonomy 2024 [\[arxiv\]](#)

Constrained LTL Specification Learning from Examples

C. Zhang*, P. Kapoor*, I. Dardik, A. Cui, R.M. Goes, D. Garlan, E. Kang

- IEEE International Conference on Software Engineering (ICSE) 2025 [\[arxiv\]](#)

Analyzing Tolerance of Reinforcement Learning Controllers against Deviations in Cyber Physical Systems

C. Zhang*, P. Kapoor*, R.M. Goes, D. Garlan, E. Kang, A. Ganlath, S. Mishra, N. Ammar

- Formal Methods (FM) 2024 [\[arxiv\]](#)

Safe Planning through Incremental Decomposition of Signal Temporal Logic

P. Kapoor, R.M. Goes and E. Kang

- Nasa Formal Methods (NFM) 2024 [\[arxiv\]](#)

Follow The Rules: Online Signal Temporal Logic Tree Search for Guided Imitation Learning in Stochastic Domains

J. Patrikar, J. Aloor, P. Kapoor, S. Scherer and J. Oh

- IEEE International Conference on Robotics and Automation (ICRA) 2023 [\[arxiv\]](#)

Challenges in Close-Proximity Safe and Seamless Operation of Manned and Unmanned Aircraft in Shared Airspace

J. Patrikar, J. Dantas, S. Ghosh, P. Kapoor et al

- IEEE International Conference on Robotics and Automation (ICRA) 2022 [\[arxiv\]](#)

Model-based Reinforcement Learning from Signal Temporal Logic Specifications

P. Kapoor, A. Balakrishnan, J. V. Deshmukh

- [\[arxiv\]](#)